

SAT 9
Math Module 3

Question # ID
SAT9 M 3.1 35bec412

Answer
79

73, 74, 75, 77, 79, 82, 84, 85, 91
What is the median of the data shown?

SAT9 M 3.2 4ff7b652

B

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

SAT9 M 3.3 a09ma303

B

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. In the table, a is a constant. What is the value of a ?

☐ A 67	☐ C 32
☐ B 35	☐ D 27

SAT9 M 3.4 5a7ab8e8

C

$66x = 66x$
How many solutions does the given equation have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. Zero

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Question #	ID		Answer
SAT9 M 3.5	c048055c	A model predicts that the population of Springfield was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$? A. $p = 0.96(15,000)^x$ B. $p = 1.04(15,000)^x$ C. $p = 15,000(0.96)^x$ D. $p = 15,000(1.04)^x$	D
SAT9 M 3.6	548a4929	The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy-plane has an x-intercept at $(a, 0)$ and a y-intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$? A. 21 B. 28 C. 32 D. 35	A
SAT9 M 3.7	52931bfa	Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$? A. $\frac{x-7}{5}$ B. $\frac{8x-3}{2}$ C. $\frac{8x^2-3x-14}{2x-14}$ D. $\frac{8x^2-3x-77}{2x-14}$	B
SAT9 M 3.8	f329442c	Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A? A. 43 B. 86 C. 129 D. 1,849	D

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Question # ID
SAT9 M 3.9 24cec8d1

Answer

46

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

SAT9 M 3.10 03503d49

B

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is \$14,000, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

- A. \$148.15
- B. \$161.53
- C. \$172.84
- D. \$184.94

SAT9 M 3.11 f811d345

113

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

SAT9 M 3.12 a09ma312

D

The positive number a is 230% of the number b , and a is 60% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

(A) 138

(B) 217

(C) 283

(D) 383

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Question # ID
SAT9 M 3.13 9b5b23fc

Answer

D

For $x > 0$, the function f is defined as follows:
 $f(x)$ equals 201% of x
Which of the following could describe this function?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

SAT9 M 3.14 3008cfc3

33

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

SAT9 M 3.15 cc7ffe02

C

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

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Question # ID
SAT9 M 3.16 a09ma316

Answer

B

$$3x = 36y - 45$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

(A) $x = 4y$

(B) $\frac{1}{3}x = 4y$

(C) $x = 12y - 15$

(D) $\frac{1}{3}x = 12y - 15$

SAT9 M 3.17 fbb96bb1

C

$$x - 29 = (x - a)(x - 29)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

I. a

II. $a + 1$

III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

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Question # ID
SAT9 M 3.18 a09ma318

Answer

D

One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S , in gallons, needed to stain the fence in this yard twice?

(A) $S = \frac{w}{170}$

(B) $S = 170w$

(C) $S = 340w$

(D) $S = \frac{w}{85}$

SAT9 M 3.19 a54753ca

C

In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$

SAT9 M 3.20 a09ma320

168

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?

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Question # ID
SAT9 M 3.21 a09ma321

Answer

A

The area of a rectangular region is increasing at a rate of 250 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

(A) 0.39

(B) 1.27

(C) 13.67

(D) 23.24

SAT9 M 3.22 a35c7164

B

$$\begin{aligned}5x + 7y &= 1 \\ ax + by &= 1\end{aligned}$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A. $\begin{aligned}10x + 7y &= 1 \\ ax - 2by &= 1\end{aligned}$

B. $\begin{aligned}10x + 7y &= 1 \\ ax + 2by &= 1\end{aligned}$

C. $\begin{aligned}10x + 7y &= 1 \\ 2ax + by &= 1\end{aligned}$

D. $\begin{aligned}5x - 7y &= 1 \\ ax + by &= 1\end{aligned}$